

```
5 : "categorical_cols"
6 : "id_cols"
7 : "text_cols"
8 : "boolean_cols"
9 : "duplicate_row_count"
10 : "data_quality_score"
11 : "missing_pct"
```

```
]
```

Step 3: Classification done

```
▼ {
  "dataset_type" :
  "subscription"
  "key_entity" : "customer"
  "id_column" : "customer_id"
  ▼ "date_columns" : [
    0 : "created_date"
    1 : "canceled_date"
  ]
  ▼ "value_columns" : [
    0 : "subscription_cost"
    1 :
    "subscription_interval"
    2 :
    "was_subscription_paid"
  ]
  "status_column" :
  "was_subscription_paid"
  ▶ "group_columns" : []
  "confidence" : 1
  "reasoning" :
  "This dataset represents
  customer subscription
  information, with the
  customer_id serving as the
  primary key and
  was_subscription_paid
  indicating whether the
```

Chat Assistant

Dataset Report



Conversational Data Assistant

Ask questions in natural language, request dataset overviews, or search for trends and anomalies.



What is this dataset about?



This is a Python script that appears to be analyzing a dataset of customer information. The data is stored in a list of dictionaries, where each dictionary represents a single customer and contains various attributes such as `customer_id`, `created_date`, etc.

The script uses the pandas library to manipulate and analyze the data. Here's a breakdown of what the script does:

- Importing necessary libraries:
 - `pandas` is used for data manipulation and analysis.
 - `datetime` is used to parse date strings into datetime objects.
- Loading the dataset:
 - The dataset is loaded from a list of dictionaries, where each dictionary represents a single customer.
- Filtering customers based on certain conditions:
 - Customers who were created on or after September 1st, 2022 are filtered out.
 - Customers with `created_date` in the format 'YYYY-MM-DD' are converted to datetime objects for easier comparison and analysis.
- Grouping customers by their creation date:
 - The script groups the customers by their creation date using the `groupby` function from pandas.
- Calculating the number of customers created on each day:
 - The script calculates the number of customers created on each day using the `size` method of the grouped data.
- Printing the results:
 - The script prints the results, showing the creation date and the number of customers created on that date.

Here's a more detailed version of the code with comments:

```
import pandas as pd
from datetime import datetime

# Load the dataset from a list of dictionaries
data = [
    # Customer 1
    {'customer_id': 1, 'created_date': '2022-09-01'},
    {'customer_id': 2, 'created_date': '2022-08-31'},
    {'customer_id': 3, 'created_date': '2022-09-02'},
    # ... and so on
]

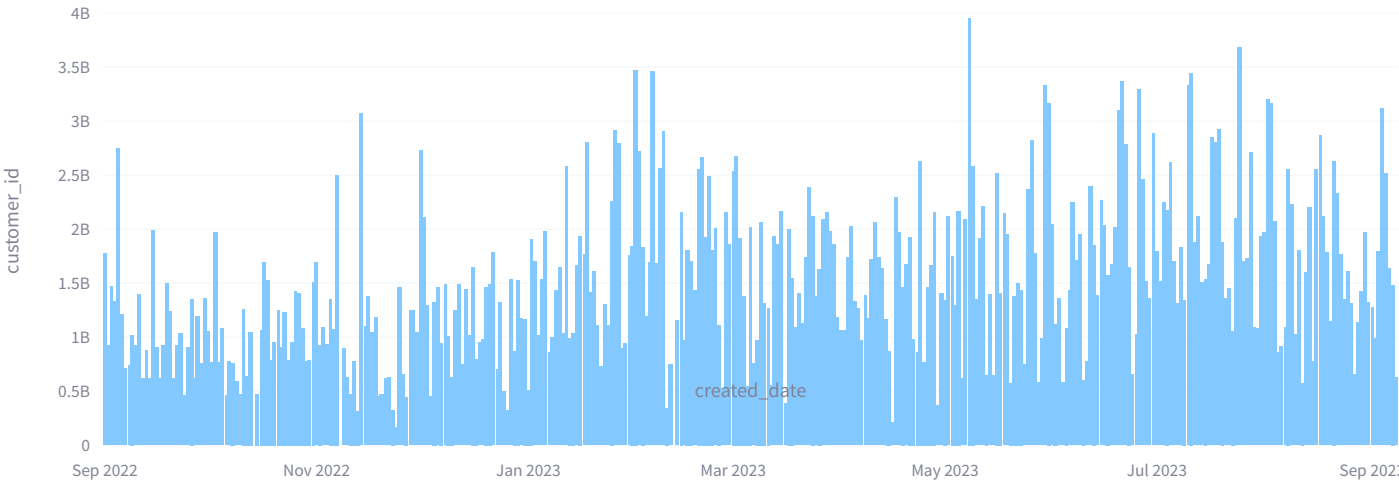
# Convert the created_date column to datetime objects
df = pd.DataFrame(data)
df['created_date'] = pd.to_datetime(df['created_date'])

# Filter out customers who were created on or after September 1st, 2022
df = df[df['created_date'] < '2022-09-01']

# Group the data by creation date and calculate the number of customers
grouped_data = df.groupby('created_date').size()

# Print the results
print(grouped_data)
```

	customer_id	created_date	canceled_date	subscription_cost	subscription_interval	was_subscription_paid
31	144,402,204	2022-09-04T00:00:00	2022-09-22T00:00:00	39	month	☒
32	154,747,678	2022-09-04T00:00:00	2023-01-15T00:00:00	39	month	☒
33	130,939,158	2022-09-04T00:00:00	2023-02-19T00:00:00	39	month	☒
34	150,912,359	2022-09-04T00:00:00	2022-11-02T00:00:00	39	month	☒
35	154,132,592	2022-09-04T00:00:00	2022-11-01T00:00:00	39	month	☒
36	153,368,891	2022-09-04T00:00:00	2022-10-11T00:00:00	39	month	☒
37	154,225,450	2022-09-05T00:00:00	None	39	month	☒
38	154,904,040	2022-09-05T00:00:00	2022-09-06T00:00:00	39	month	☒
39	127,604,200	2022-09-05T00:00:00	2022-10-06T00:00:00	39	month	☒
40	154,913,576	2022-09-05T00:00:00	2023-03-13T00:00:00	39	month	☒
41	154,910,432	2022-09-05T00:00:00	2023-03-13T00:00:00	39	month	☒



What are the columns in the dataset?



The data appears to be a list of dictionaries, where each dictionary represents a customer's subscription information. The keys in the dictionary are likely to represent different attributes of the subscription, such as `customer_id` , `subscription_cost` , and `subscription_interval` .

To analyze this data, we can use various statistical and machine learning techniques. Here are some potential steps:

1. **Data cleaning:** Before analyzing the data, we should check for any missing or duplicate values. We can also convert the `subscription_cost` column to a numerical value if it's currently stored as a string.
2. **Summary statistics:** Calculate summary statistics such as mean, median, and standard deviation of the subscription costs to understand the distribution of costs.
3. **Correlation analysis:** Analyze the correlation between different attributes, such as `customer_id` , `subscription_cost` , and `subscription_interval` . This can help identify potential relationships or patterns in the data.
4. **Machine learning models:** Train machine learning models, such as regression or classification models, to predict subscription costs based on other attributes.

Here's an example of how we could use Python and the Pandas library to analyze this data:

```
import pandas as pd

# Load the data into a Pandas DataFrame
df = pd.read_json('subscription_data.json')

# Check for missing values
print(df.isnull().sum())

# Convert subscription_cost column to numerical value
df['subscription_cost'] = df['subscription_cost'].astype(float)

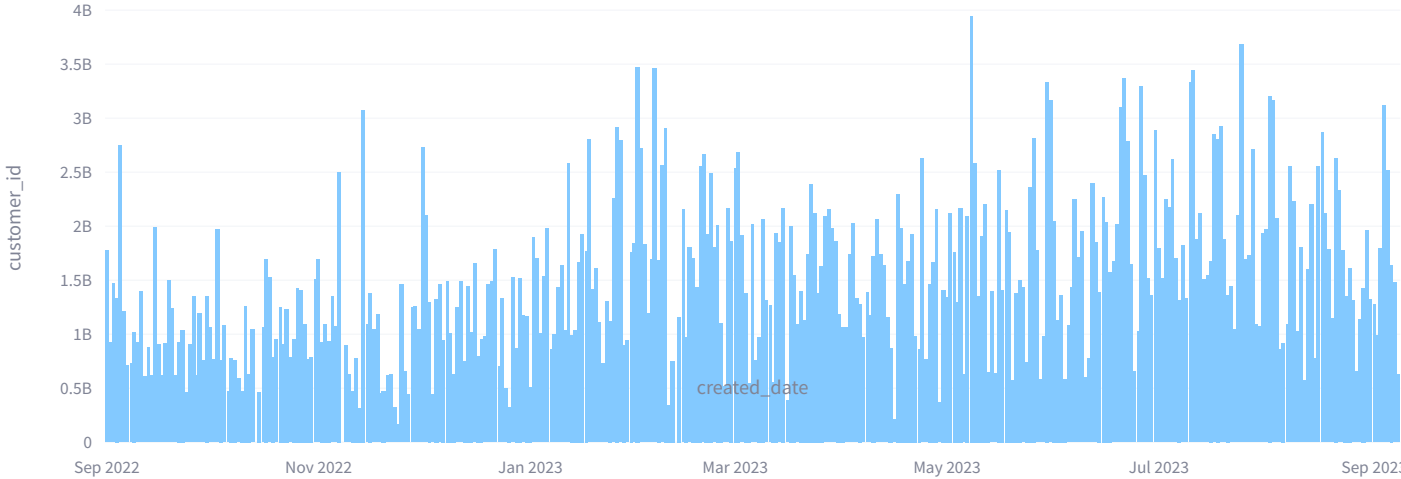
# Calculate summary statistics
mean_cost = df['subscription_cost'].mean()
median_cost = df['subscription_cost'].median()
std_dev = df['subscription_cost'].std()

print(f'Mean subscription cost: {mean_cost:.2f}')
print(f'Median subscription cost: {median_cost:.2f}')
print(f'Standard deviation of subscription costs: {std_dev:.2f}')

# Analyze correlation between attributes
corr_matrix = df[['customer_id', 'subscription_cost', 'subscription_interval']].corr()
print(corr_matrix)
```

This is just a starting point, and there are many other techniques we could use to analyze this data. Do you have any specific questions or areas of interest?

	customer_id	created_date	canceled_date	subscription_cost	subscription_interval	was_subscription_paid
40	158,849,640	2022-10-23T00:00:00	None	39	month	☒
41	159,276,280	2022-10-26T00:00:00	None	39	month	☒
42	153,255,183	2022-11-01T00:00:00	2023-02-16T00:00:00	39	month	☒
43	159,620,660	2022-11-02T00:00:00	2022-12-12T00:00:00	39	month	☒
44	160,243,008	2022-11-03T00:00:00	2022-12-29T00:00:00	39	month	☒
45	155,334,332	2022-11-05T00:00:00	2022-12-04T00:00:00	39	month	☒
46	160,523,170	2022-11-05T00:00:00	2022-12-12T00:00:00	39	month	☒
47	156,078,158	2022-11-06T00:00:00	2023-01-25T00:00:00	39	month	☒
48	136,157,937	2022-11-06T00:00:00	2022-12-13T00:00:00	39	month	☒
49	138,919,619	2022-11-07T00:00:00	2022-12-06T00:00:00	39	month	☒
						☐





What is the average subscription cost?



The average subscription cost is \$39.00.

↓	avg(subscription_cost)
0	39

Ask a question about your data...

